

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE CODE: ITA 06

COURSE NAME: MACHINE LEARNING

COURSE OUTCOME COVERED

CO1: Learning Problems, Perspectives and Issues, Concept Learning, Version Spaces & Candidate Elimination, Inductive Bias, Decision Tree Learning, Representation & Heuristic Search (BL1, BL2)

Assessment Tool Weightage: 10%

QUESTIONS: 15

Total Marks: 25

Annexure A

TECHNICAL PRESENTATION

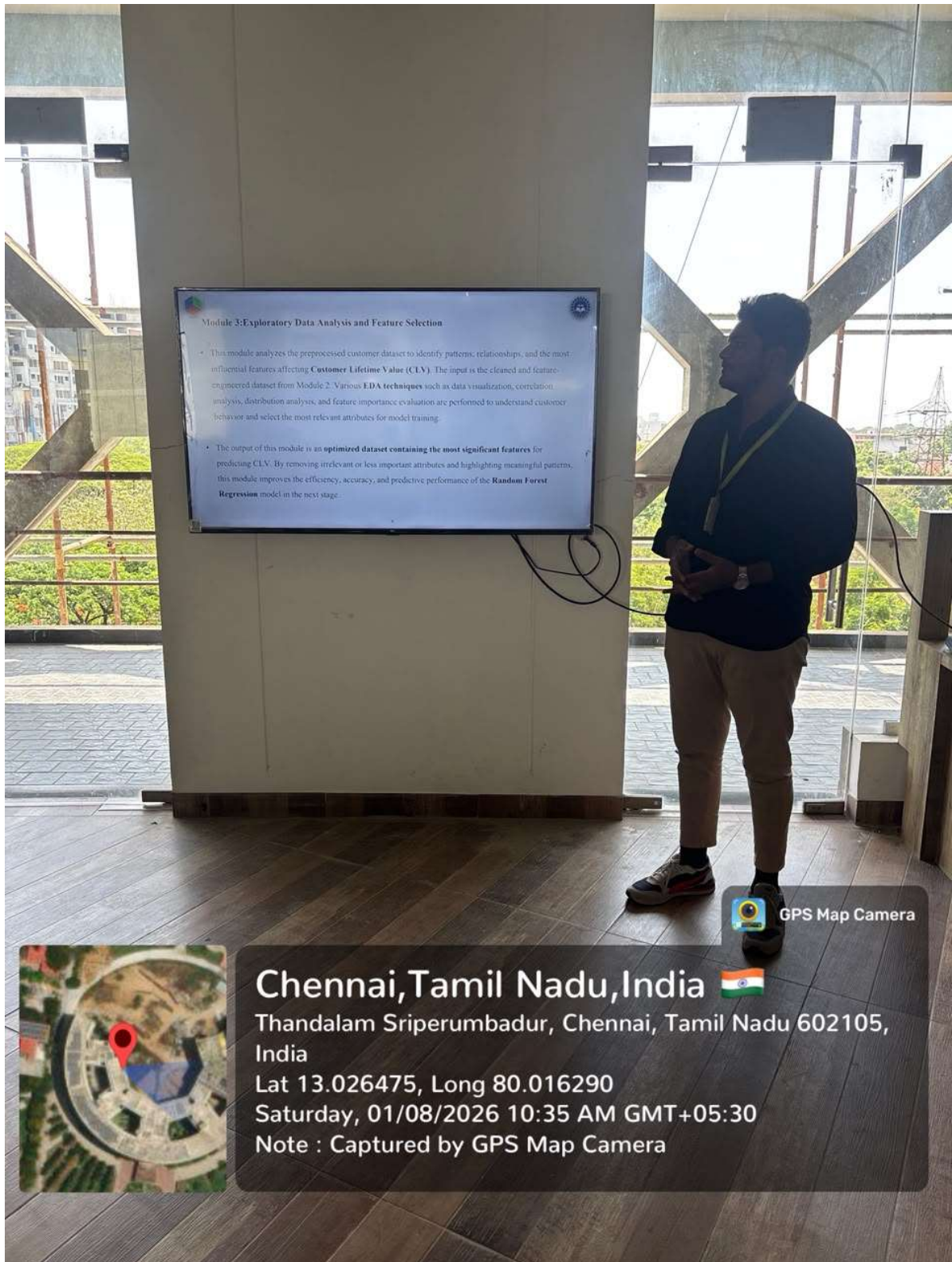
Code of Conduct:

I Anirudhan S / 192312056 certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

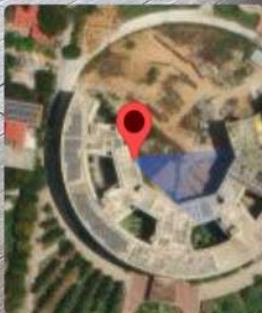
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Anirudhan S



GPS Map Camera



Chennai, Tamil Nadu, India 🇮🇳

Thandalam Sriperumbadur, Chennai, Tamil Nadu 602105, India

Lat 13.026475, Long 80.016290

Saturday, 01/08/2026 10:35 AM GMT+05:30

Note : Captured by GPS Map Camera

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Conceptual Understanding	35	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	20	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application of Knowledge	25	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	10	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	10	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand




SIMATS ENGINEERING
Saveetha Institute of Medical and Technical Sciences
Chennai-602105.

Customer Lifetime Value Prediction Using Machine Learning Regression

K.Kameshwaran 192312046 Deepan Kumar P 192312498 Anirudhan S 192312056 Shaik Arshad Basha 192312336 Thiruvengadaprasath S 192312016	Supervisor Dr K Baskar, Professor Department of Computer Science and Engineering SIMATS Engineering
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1




Abstract

- This project aims to predict Customer Lifetime Value (CLV) using Random Forest Regression to help businesses identify high-value customers and improve decision-making. The model uses customer data such as Age, Gender, Purchase Frequency, Total Purchase Amount, Average Order Value, Recency, and Total Revenue collected from a Kaggle dataset.
- The methodology includes data collection, preprocessing, feature engineering, exploratory data analysis (EDA), model training, and evaluation using MAE, RMSE, and R² Score. The predicted CLV is used to classify customers into Gold, Silver, and Bronze segments and is displayed through a dashboard with business recommendations.
- The system provides accurate CLV predictions, enabling businesses to improve customer retention, optimize marketing strategies, and make data-driven decisions.

2




Problem Statement:

- Many businesses focus on short-term sales instead of understanding the **long-term value of their customers**, making it difficult to identify high-value customers and develop effective retention strategies. Traditional methods of estimating Customer Lifetime Value (CLV) are often inaccurate and time-consuming.
- This project addresses this problem by using **Machine Learning (Random Forest Regression)** to accurately predict Customer Lifetime Value based on customer purchasing behavior and demographic data. The predicted CLV helps businesses segment customers, improve retention, optimize marketing efforts, and make data-driven business decisions.

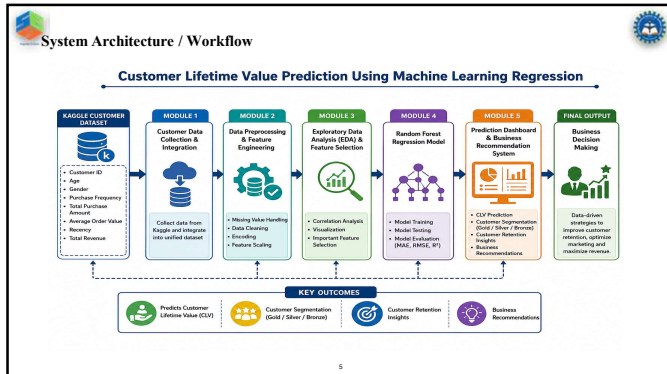
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Objectives

- To design and implement a robust Random Forest Regression model capable of accurately predicting Customer Lifetime Value (CLV) by analyzing customer demographics, purchasing behavior, and transaction history, enabling precise long-term revenue forecasting and strategic business planning.
- To develop an intelligent customer segmentation and recommendation system that classifies customers into Gold, Silver, and Bronze categories, delivers personalized retention strategies, optimizes marketing resource allocation, increases customer loyalty, and maximizes overall business growth and profitability through data-driven decision-making.

4



5

Module 1: Customer Data Collection and integration

- This module focuses on collecting customer information from the **Kaggle dataset** and integrating it into a unified, structured format. The input includes customer details such as **Customer ID, Age, Gender, Purchase Frequency, Total Purchase Amount, Average Order Value, Recency, and Total Revenue**. These attributes provide the foundation for analyzing customer behavior and predicting Customer Lifetime Value (CLV).
- The output of this module is a complete and well-organized customer dataset that is ready for the **Data Preprocessing and Feature Engineering** stage. By ensuring data consistency and proper integration, this module establishes a reliable foundation for accurate machine learning model training, customer segmentation, and business recommendation generation.

6

Module 2: Data Preprocessing and Feature Engineering

- This module prepares the integrated customer dataset for machine learning by cleaning and transforming the data into a suitable format. The input is the structured dataset generated from Module 1, containing customer attributes such as **Age, Gender, Purchase Frequency, Total Purchase Amount, Average Order Value, Recency, and Total Revenue**. Data preprocessing techniques including **missing value handling, duplicate removal, categorical encoding, feature scaling, and feature engineering** are applied to improve data quality and model performance.
- The output of this module is a **clean, normalized, and feature-enhanced dataset** that is ready for **Exploratory Data Analysis (EDA)** and feature selection. By eliminating inconsistencies and creating meaningful features, this module improves the accuracy, reliability, and efficiency of the Random Forest Regression model used for Customer Lifetime Value prediction.

7

Module 3: Exploratory Data Analysis and Feature Selection

- This module analyzes the preprocessed customer dataset to identify patterns, relationships, and the most influential features affecting **Customer Lifetime Value (CLV)**. The input is the cleaned and feature-engineered dataset from Module 2. Various **EDA techniques** such as data visualization, correlation analysis, distribution analysis, and feature importance evaluation are performed to understand customer behavior and select the most relevant attributes for model training.
- The output of this module is an **optimized dataset containing the most significant features** for predicting CLV. By removing irrelevant or less important attributes and highlighting meaningful patterns, this module improves the efficiency, accuracy, and predictive performance of the **Random Forest Regression** model in the next stage.

8




Module 4: Customer Lifetime value Prediction using Random Forest Regression

- This module develops and trains the **Random Forest Regression** model to predict **Customer Lifetime Value (CLV)** using the optimized dataset obtained from Module 3. The input consists of the selected customer features, including **Age, Gender, Purchase Frequency, Total Purchase Amount, Average Order Value, Recency, and Total Revenue**. The model is trained and tested, and its performance is evaluated using **Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and R² Score** to ensure accurate and reliable predictions.
- The output of this module is the **predicted Customer Lifetime Value (CLV)** for each customer along with a validated machine learning model. These predictions are then forwarded to the **Prediction Dashboard and Business Recommendation System**, where customers are segmented into **Gold, Silver, and Bronze** categories to support business decision-making and customer retention strategies.

9




Module 5: Prediction Dashboard and Business Recommendation System

- This module presents the predicted **Customer Lifetime Value (CLV)** in an interactive dashboard and converts the prediction results into meaningful business insights. The input consists of the predicted CLV values generated by the **Random Forest Regression** model. Based on these predictions, customers are classified into **Gold, Silver, and Bronze** segments, and key metrics are visualized through charts, graphs, and summary reports to simplify analysis.
- The output of this module is a **prediction dashboard** displaying CLV results, customer segments, retention insights, and personalized business recommendations. This module enables businesses to identify high-value customers, improve customer retention strategies, optimize marketing campaigns, and make informed, data-driven decisions that enhance long-term profitability and customer satisfaction.

10




Conclusion & Future Work

- This project successfully predicts **Customer Lifetime Value (CLV)** using **Random Forest Regression** based on customer demographic and purchase data. The system accurately segments customers into **Gold, Silver, and Bronze** categories and provides business recommendations, helping organizations improve customer retention and make data-driven decisions.
- The system can be enhanced by integrating **real-time customer data**, implementing advanced models such as **XGBoost** and **Deep Learning**, and adding features like **customer churn prediction**, personalized recommendations, and cloud-based deployment for improved scalability and accuracy.

11




References

- [1] Y. Yan and N. Resnick, "A Data-Driven Approach to Customer Lifetime Value Prediction Using Probability and Machine Learning Models," *Data Analytics Journal*, vol. 3, Art. no. 100601, Elsevier, 2025.
- [2] L. Taherkhani, A. Daneshvar, H. Amoozad Khalili, and M. R. Sanaei, "Analysis and Optimization of Customer Lifetime Value Prediction Using Machine Learning and Deep Learning Models by RFM Techniques," *International Journal of Web Research*, vol. 8, no. 2, pp. 79–92, Apr. 2025.
- [3] "Machine Learning for Predicting Customer Lifetime Value in Retail," *Canadian Journal of Marketing Research*, vol. 14, no. 3, Oct. 2024.
- [4] Y. Yan and N. Resnick, "A High-Performance Turnkey System for Customer Lifetime Value Prediction in Retail Brands," *Quantitative Marketing and Economics*, vol. 22, pp. 169–192, 2024.

12